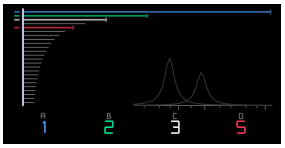
MUSICA OBSCURA

six modules for the original OP-1 (op1r): two synths, two effects, two sequencers each built on an obscure or half-forgotten idea from the history of sound



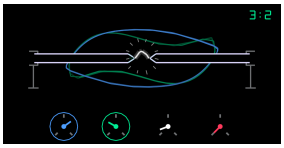
SYNTH

mixtur

the undertone organ

Nineteenth-century theorists argued about a ghost: the undertone series f, f/2, f/3 … mirroring the overtones, which nothing in nature produces. In the 1950s Oskar Sala made it real with the Mixtur-Trautonium: frequency dividers sounded the undertones of every note, coloured by formant circuits. It is the sound of the bird attacks in Hitchcock's The Birds (1963).

- The key you play is the **top** note; four phase-locked saws sound f/n beneath it.
- **blue green white orange**: undertone n of divider A–D (off, 1–24). Default 1·2·3·5 = a minor chord below the key.
- **Press** for page 2: **blue** formant 1 · **green** formant 2 · **white** resonance (vowel colour).
- Screen: the undertone ladder; string n is 1/n long, beads step down the active strings.



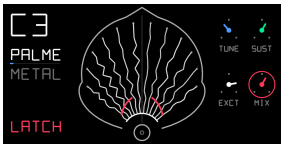
SYNTH

arc

Duddell's singing street lamp

London, 1899: carbon arc street lamps hissed and howled. William Duddell found the arc has negative resistance; shunted with a capacitor and an inductor it became an oscillator, and he played tunes on it from a keyboard: one of the first electronic instruments. Each voice is a Van der Pol oscillator, the textbook model of exactly this circuit.

- **blue** current: sine → reedy buzz → screaming relaxation, always in tune.
- **green** carbon (asymmetry, even harmonics) · **white** hiss (plasma noise, pitch wander) · **orange** strike + sputter (ignition swoop, flicker while held).
- **Press** for page 2: **blue** 2nd lamp ratio (½–2, detents) · **green** coupling: off → lock/beat → chaos · **white** brightness.
- Screen: the live phase portrait, a circle that bends into the relaxation loop; the 2nd lamp in green.



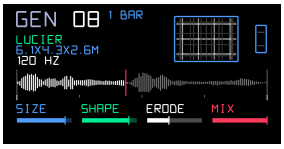
FX

palme

the Ondes Martenot string speaker

The Ondes Martenot (1928; Messiaen, Jonny Greenwood) played through diffuseurs. The **Palme** (1950) is a palm-leaf soundboard with 12 strings tuned chromatically that ring in sympathy with every note; the **Métallique** replaced the speaker cone with a gong. Ondistes lift the dampers to let it ring on.

- Works on any track FX or the master. Each note you play sets its string ringing.
- **blue** root note (C2–B4) · **green** sustain (0.25–30 s) · **white** excite · **orange** mix.
- **orange press**: LATCH, étouffoir off. Whatever rings becomes a drone; press again to let it fade.
- **Press blue green white** for page 2: métallique (palme → gong) · damping colour · detune spread.



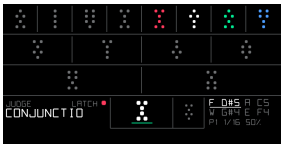
FX

in a room

I am sitting in a room

In 1969 Alvin Lucier recorded himself speaking, played the tape back into the room and recorded it again, generation after generation, until the words were gone and only the room's resonances were left. Here the room is modelled from real dimensions (axial modes f = n·c/2L) and every pass of a loop is one generation.

- **orange press** = record a phrase, then GEN 01, 02 … counts up. Best on the **master** FX.
- **blue** size (0.8–8 m) · **green** shape: CUBE, GOLDEN, BOLT, LUCIER · **white** erode · **orange** mix.
- **Press blue** for page 2: loop FREE 0.5–3.9 s or synced 1/16 … 2 bars (**orange** press then arms and records from the next beat / downbeat) · **press green**: absorb.
- Tip: small CUBE or GOLDEN rooms give clear chords; the real Lucier room is very low on the speaker.



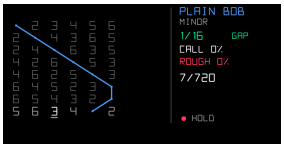
SEQUENCER

oracle

the geomancer's drum machine

Geomancy, ilm al-raml ("the science of sand"), came from the Arab world to medieval Europe (Agrippa, Fludd). Four random **Mothers** (4 rows of one or two dots) generate Daughters, Nieces, Witnesses, a **Judge** and a Reconciler by fixed rules. 16 figures × 4 rows = a 16-step, 4-lane pattern from 16 random bits; the Judge can only ever be 8 of the 16 figures.

- Shift + sequencer key: choose **oracle**. **Hold a key** to play it transposed; Shift + key latches; Shift + C# start / stop.
- **blue green white orange**: figure of Mother 1–4 (named). **Press** an encoder: re-cast that Mother.
- Shift + **orange** press: voicing (open, major, minor, sus4, 7ths, quartal …) · Shift + **white** press: EDIT lanes (keys set interval / drum) · Shift + **white** turn: lane.
- Shift + **blue** rate · Shift + **green** gate · Shift + F G A B patterns · Shift + G A B C mute lanes. Drums: any key starts, lanes keep their drum sounds.



SEQUENCER

peal

English change ringing

English ringers do not play tunes: since the 1600s (Stedman, Campanalogia, 1677) they permute the order of the bells row by row by fixed methods, never repeating a row until the bells come back to rounds. All 40,320 orders of 8 bells were rung once, in 1963, in about 18 hours. Ringers learn their bell's path, the blue line.

- Shift + sequencer key: choose **peal**. **Hold 3–8 keys**: they ring while held (highest = treble).
- Shift + keys latch · Shift + C# start / stop. **blue** method (Plain Hunt, Plain Bob, Grandsire, Stedman), press = handstroke gap.
- **green** rate, press = "That's all" (stop at rounds) · **white** auto-call chance, press = Bob (Shift: Single).
- **orange** rough striking, press = back to rounds · Shift + **blue green white orange**: octave · gate · level · start/stop.

Encoders left to right: blue, green, white, orange. "Press" = push the encoder. Resource monitor: Shift + help, then T1. Flash at your own risk; hold COM at boot to update or go back. Test build: plugin numbers are provisional, so use throwaway patches. Built on op1r and its plugin SDK by popcorp (pcppcp). · ultralazr, 2026